

Plackett-Burman Designs

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Introduction

Plackett–Burman (PB) designs, introduced by Plackett and Burman in 1946, are two-level orthogonal screening designs with n runs (where n is a multiple of 4) that can accommodate up to $n - 1$ factors. They are widely regarded as the most economical screening tool in industrial DoE: when many candidate factors must be sifted to identify the few that drive the response, a 12-run PB design can screen up to 11 factors, a 20-run PB up to 19, and so on. The price for this economy is a resolution-III aliasing structure in which main effects are confounded with two-factor interactions, so PB designs are appropriate strictly for screening — never for confirmatory or interaction-aware inference.

Prerequisites

Familiarity with fractional factorial designs, the concept of resolution, the effect-sparsity principle, and the distinction between regular and non-regular fractional factorial constructions.

Theory

PB designs are built from Hadamard matrices, square matrices of ± 1 entries with mutually orthogonal columns. With n runs the design supplies $n - 1$ orthogonal contrast columns, each of which can carry one factor. For $n = 12$, 11 factors fit; for $n = 20$, 19 factors.

Non-regular PB designs (those whose run count is a multiple of 4 but not a power of 2, like 12, 20, 24) have a famously complex partial-aliasing structure: each main effect is aliased with multiple two-factor interactions at fractional weights, rather than fully confounded with a single one. This complex partial aliasing is a property the experimentalist must understand to interpret active-effect candidates correctly. Regular PB designs (a power of 2) coincide with standard resolution-III fractional factorials and have cleaner alias chains.

Assumptions

Two-factor and higher-order interactions are negligible compared with main effects (the sparsity-of-effects principle), and follow-up experiments will be conducted on the screened-in subset to characterise interactions properly. PB designs are a tool of two-stage industrial DoE: screen first, characterise second.

R Implementation

```
library(FrF2)

pb <- pb(nruns = 12, nfactors = 8, randomize = TRUE, seed = 2026)
pb

for (v in paste0("A", 1:8)) {
  pb[[v]] <- as.numeric(as.character(pb[[v]]))
}
set.seed(2026)
pb$y <- 10 + 2.5 * pb$A1 + 1.8 * pb$A2 + rnorm(nrow(pb), 0, 0.6)
```

```
fit <- lm(y ~ ., data = pb)
summary(fit)
```

Output & Results

`pb()` constructs a randomised Plackett–Burman layout for the requested run count and factor count. The main-effects regression returns coefficients for each factor; combining the regression with a Lenth-style or half-Normal-plot identification of the active subset is the standard analytical approach for unreplicated PB designs.

Interpretation

A reporting sentence: “The 12-run Plackett–Burman design screened eight two-level factors, identifying A1 (effect 2.5, $p < 0.001$) and A2 (effect 1.8, $p < 0.001$) as the principal drivers of the response. The remaining six factors had effect estimates within Lenth’s pseudo-standard-error envelope and were dropped from follow-up. A subsequent 2^2 full factorial with replicated centre points characterised curvature and the A1 $\times$ A2 interaction in the active subset.” Always describe both active and inactive factors and the screening criterion used.

Practical Tips

- Plackett–Burman is a screening design; always follow it up with a higher-resolution fractional factorial, full factorial, or response-surface design on the active factor subset. Drawing causal conclusions directly from a PB result without follow-up is methodologically unsound.
- The 12-run design is the most popular member of the family because it balances economy (one-third the runs of a comparable resolution-IV fractional factorial) with the orthogonality and ease-of-interpretation of regular Hadamard-based construction.
- Complex partial aliasing in non-regular PB designs can occasionally reveal active two-factor interactions indirectly through effect-heredity logic, but this should be regarded as exploratory; formal interaction inference still requires a higher-resolution follow-up design.
- Always randomise run order to protect against temporal confounding; running treatments in design order conflates time-trend effects with factor effects in any unreplicated screening design.
- Report transparently which factors were screened in and out of the follow-up; the pruning decision is itself a substantive analytical choice and reviewers will want to see the criterion used.
- For mixed-level screening, definitive screening designs (DSDs, Jones and Nachtsheim 2011) generalise PB ideas to three-level factors and offer cleaner analysis of curvature alongside main-effect screening.

R Packages Used

`FrF2::pb()` for Plackett–Burman construction with explicit randomisation; `DoE.base::oa.design()` for general Hadamard-matrix-based orthogonal arrays; `daewr` for textbook companion examples and analysis helpers; `unrep` for Lenth-style identification of active effects in unreplicated PB designs; `dss` and `daewr::DefScreen()` for definitive screening designs as a modern alternative.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Observational designs and STROBE
- Randomised controlled trials
- Adaptive, non-inferiority, equivalence trials
- Bench and translational design
- MCAR, MAR, MNAR
- Multiple imputation with mice
- Linear mixed models with lme4
- GLMMs and GEE
- DAGs with dagitty and ggdag
- Propensity scores and IPTW
- G-methods, IV, DiD, RDD; heterogeneity of treatment effect
- SIR and SEIR models with deSolve
- Pre-registration and statistical analysis plans